

Project Plan — Signalpost Norway Agent

Challenge window: **Aug 23 – Oct 21, 2026** (59 days). Up to 4 revisions allowed before Oct 18, each frozen before its next daily run.

Guiding principle

Precision beats coverage. The hard gates require 95% external precision and disqualify on wrong-company publication or fabricated financials — so every phase below treats "don't claim something we can't support" as a first-class requirement, not an afterthought.

Phase 0 — Setup (Day 0–1)

- Create private GitHub repo, push this scaffold
- Download full 411,160-company universe (JSONL) + published hashes, verify hash
- Download runnable starter agent, read end-to-end once
- Read all brief docs: challenge brief, output contract, source policy, agent playbook, evaluation contract
- Set up Python env, pin dependency versions in `requirements.txt`
- Pick a **dev slice**: ~200–300 companies sampled from the universe to iterate against locally (never the live daily 100)

Phase 1 — Entity resolution (Day 2–7)

- Pydantic model for canonical entity (org number, legal name, brand name(s), registered address, status)
→ see `docs/data_schema.md`
- Registry lookup by org number → confirm entity exists / is active
- Official-site discovery: search + heuristics (domain contains legal name tokens, registry-listed URL if present, .no TLD preference)
- Confidence scoring for "is this really their site" before anything downstream trusts it — this is the single highest-leverage correctness check in the whole system

Phase 2 — Crawl + extract (Day 7–20)

- Scrapy spider skeleton: sitemap-first crawl, depth-limited fallback
- extract for JSON-LD / microdata / OpenGraph on company pages
- Trafilatura for clean text extraction (about pages, news, press)
- Playwright fallback **only** for JS-only sites (keep it as last resort — it's slow and burns the request/time budget fast)
- LinkedIn company page discovery + basic public-field extraction (headcount band, industry, described locations)
- Job-board signal discovery (is the company actively hiring — yes/no + count if available, not scraped listings content)
- Respect docs/source_policy.md boundaries throughout

Phase 3 — Matching (Day 15–25, overlaps Phase 2)

- RapidFuzz-based candidate matching: entity name variants vs. found candidates (site title, LinkedIn name, news mentions)
- Reject-below-threshold logic — an unresolved field must stay **missing**, never a low-confidence guess
- Distinguish brand vs. legal entity vs. subsidiary vs. parent explicitly in the schema (this is called out directly in the brief as a common failure mode)

Phase 4 — Provenance + validation (Day 20–30)

- Every claim carries: source URL, retrieval timestamp, reporting period (where relevant), availability state (found / not_applicable / blocked / missing — never silently zero)
- Pydantic validators enforce the output contract before anything is written to data/processed/
- Reject-and-log path for anything that fails validation, instead of writing partial/incorrect records

Phase 5 — Refresh + idempotency (Day 25–35)

- Snapshot storage design: append-only dated snapshots per company, previous snapshot always preserved

- Diff logic: re-run on same company → detect and surface *meaningful* changes only (new filing, new role, new location, new job signal)
- Idempotency test: run the same batch twice, confirm no duplicate records and identical output when nothing changed upstream

Phase 6 — Learning harness (Day 20–40, overlaps 3–5)

- Scorer for dev-slice runs: exact-company-match rate, supported-field rate, bad-claim rate, runtime, request count, declared cost
→ see docs/learning_harness.md
- Strategy promotion rule: keep a new crawl/match strategy only if it adds coverage **without** increasing wrong-company or unsupported-claim rates
- Freeze discipline: lock routes/prompts/thresholds before each daily cutoff; the random 100-company run is evaluation, not training data

Phase 7 — Budget compliance (ongoing, hard checkpoint Day 30)

- Instrument every run against: 45 min wall-clock, 2,000 requests, \$10 declared third-party API spend per 100-company batch
→ see docs/api_and_budget.md
- Cache layer so repeated lookups (shared parents, common domains) don't burn request budget twice

Phase 8 — UX / dashboard (Day 35–45)

- Reuse FastAPI + React pattern from the earlier trading-agent build
- Profile browser: search, compare, "what changed since last snapshot," and a way to see *why* a field is missing (blocked vs. not found vs. not applicable)
- Desktop + mobile responsive (5% of score, keep it lightweight — don't let this phase eat time from correctness work)

Phase 9 — Scale-up run (Day 40–50)

- Move from dev slice (200–300 companies) to full submitted run (1,000+ minimum; target 10,000+ if budget/time allow)

- Full idempotent re-run to confirm nothing breaks at scale
- Confirm exactly 100 terminal result envelopes are produced per simulated daily batch (hard gate)

Phase 10 — Freeze, submit, iterate (Day 50–59)

- First freeze + submission well before Oct 18 to leave room for revisions (4 allowed)
- Watch daily 100-company scores, diagnose failures against `docs/scoring_and_gates.md`
- Use remaining revisions for targeted fixes only — not new features this late

Milestone checkpoints

Date (approx)	Milestone
Day 1	Repo + data + starter agent downloaded, read
Day 7	Entity resolution working on dev slice
Day 20	Crawl + extract producing raw candidate data on dev slice
Day 30	Provenance-complete, budget-instrumented pipeline
Day 35	Idempotent refresh verified
Day 45	Dashboard usable end-to-end
Day 50	First full-scale (1,000+) run + first submission
Day 59 (Oct 18)	Final freeze deadline
Oct 21	Challenge closes

Risks to watch

- **Wrong-company matches** are the costliest failure mode (precision gate + explicit "worse than an honest gap" framing) — bias every ambiguous decision toward missing, not a guess.
- **Playwright overuse** will blow the 45-min/2,000-request budget fast — reserve it, don't default to it.
- **LinkedIn/job-board scraping** sits closest to source-policy limits —

confirm allowed methods in docs/source_policy.md before building here.